

Simpson's Paradox Shuffle Classroom Run Kit

TOK Cognitive Science Activity Bank • Mathematics • 35-45 min

Category	Mathematics	Time	35-45 min
Difficulty	Medium	ID	simpsons-paradox-shuffle

Big idea: Aggregated data can hide subgroup structure; mathematical knowledge depends on how cases are grouped.

One-Minute Teacher Brief

Students sort subgroup data cards and discover that an overall trend can reverse when data is grouped differently.

Student Prompt

Inspect Simpson's Paradox Shuffle Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Simpson's Paradox Shuffle Stimulus A	Student-facing stimulus 1: Method A beats Method B overall, but Method B performs better inside each difficulty subgroup. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Simpson's Paradox Shuffle Stimulus B	Student-facing stimulus 2: Headline card: 'Method A has the higher success rate' beside subgroup table cards. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Simpson's Paradox Shuffle Reveal / Twist	Student-facing stimulus 3: Headline card: 'Method A has the higher success rate' beside subgroup table cards. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Aggregated data can hide subgroup structure; mathematical knowledge depends on how cases are grouped.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Simpson's Paradox Shuffle Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Simpson's Paradox Shuffle Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.

#	Question for students	Teacher key / cue
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	How do grouping choices shape statistical knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how aggregated data can hide subgroup structure; mathematical knowledge depends on how cases are grouped.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: How do grouping choices shape statistical knowledge?

Setup Checklist

- Use small whole-number subgroups so the reversal is visible without calculators.
- Ask students for the headline claim before revealing subgroup structure.

Example Stimuli or Materials

- Method A beats Method B overall, but Method B performs better inside each difficulty subgroup.
- Headline card: 'Method A has the higher success rate' beside subgroup table cards.

Resources To Use

- Resource pack: <resources/simpsons-paradox-shuffle/index.html>
- Card deck CSV: <resources/simpsons-paradox-shuffle/cards.csv>
- Visual prompt: <resources/simpsons-paradox-shuffle/visual-prompt.svg>
- Subgroup sorting mat: Group 1, Group 2, overall total.
- Responsible headline checklist: sample size, subgroup balance, denominator, and audience.

Run Sequence

1. Give groups cards for two methods split across two subgroups.
2. Students calculate or compare success rates inside each subgroup.
3. Groups combine the cards and calculate the overall rate.

4. Reveal the reversal and ask which claim is more responsible.
5. Debrief why a mathematically accurate headline can still mislead.

Debrief Moves

- Knowledge question: Can correct calculations produce a misleading knowledge claim?
- When should data be aggregated, and when should it be separated?
- How do categories and grouping decisions shape mathematical evidence?

Teacher Quick Script

- The calculation is correct; now ask whether the comparison is fair.
- What vanished when the data was combined?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...